Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. *Semester- II* *Time: 03 Hours*

Code: 101202 *Subject: Mathematics-II (Differential Equations)* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 *Answer any seven question of the following:* **[2 x 7 = 14]**

(a) Determine the type of the following differential equation

$\frac{d^2y}{dx^2} + \sin(x+y) = \sin x$

(i) Linear, homogeneous
(ii) Non-linear, homogeneous
(iii) Linear, non-homogeneous
(iv) Non-linear, non-homogeneous

(b) The value of $\Delta tan^{-1}x$ is

(i) $\Delta tan^{-1}\left(\frac{h}{1+hx+x^2}\right)$
(ii) $\Delta tan^{-1}\left(\frac{h+2x}{1+hx+x^2}\right)$
(iii) $\Delta tan^{-1}\left(\frac{h-2x}{1+hx+x^2}\right)$
(iv) $\Delta tan^{-1}\left(\frac{h+2x}{1-hx-x^2}\right)$

(c) The value of $\int_c \frac{3z^2+7z+1}{z+1}dz$ where C is $|z| = 1$ is

(i) $2\pi i$
(ii) 0
(iii) πi
(iv) $\pi i/2$

(d) If $f(z) = u(x,y) + iv(x,y)$ is an analytic function, then $f'(z) =$

(i) $\frac{\partial u}{\partial x} - i\frac{\partial v}{\partial x}$
(ii) $\frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$
(iii) $\frac{\partial v}{\partial y} - i\frac{\partial v}{\partial x}$
(iv) $\frac{\partial u}{\partial x} + i\frac{\partial u}{\partial y}$

(e) The iterative formula by using the Newton-Rapshon method for the real root of the equation $x^2 + psinx = 0$ is given by

$x_{n+1} = x_n - \frac{x^2+psinx}{qx+4cosx}$

Then the value of [p,q] is

(i) (3, 2)
(ii) (4, 3)
(iii) (4, 2)
(iv) (4, 1)

(f) The order of the rate of convergence of Newton-Raphson method is

(i) 1
(ii) 2
(iii) 3
(iv) 4

(g) Verify for exactness the differential equation:

$$y\,exp(xy).dx + [xexp(xy) + 2y]dy = 0$$

(h) State the criterion for linearly independent solutions of the homogeneous linear nth order differential equation.

© Apna BEU
CONFIDENTIAL DOCUMENT
Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

(i) Explain ordinary and singular points of a differential equation of the form:

$a_2(x)y''+a_1(x)y'+a_0(x)y=0$

(j) Find the particular integral of $(4D^2+4D-3)y=e^{2x}$, where $D\equiv\frac{d}{dx}$

Q.2 Describe the Runge-Kutta method of fourth order for the solution of initial value problem. Given the initial value problem $y'=1+y^2$, $y(0)=0$. Find $y(0{\cdot}6)$ by using Runge-Kutta method of fourth order by taking $h=0{\cdot}2$ [14]

Q.3 Find the D'Alembert's solution of the wave equation [14]

$$\frac{\partial^2 y}{\partial t^2}=c^2\frac{\partial^2 y}{\partial x^2}$$

Q.4 The points of trisection of a string are pulled aside through the same distance on opposite sides of the position of equilibrium and the string is released from rest. Derive an expression for the displacement of the string at subsequent time and show that the midpoint of the string always remains at rest. [14]

Q.5 Solve the following differential equations: [7x2=14]

(a) $2ydx+x(2logx-y)dy=0$

(b) $y-2px=tan^{-1}(xp^2)$

Q.6 (a) Solve by method of variation of parameters [7]

$$\frac{d^2y}{dx^2}+y=\tan x$$

(b) Solve by changing the independent variable [7]

$$\frac{d^2y}{dx^2}-\frac{1}{x}\frac{dy}{dx}+4x^2y=x^4$$

Q.7 (a) The following values of the function $f(x)=sinx+cosx$, are given [7]

x	:	10°	20°	30°
$f(x)$	:	1·1585	1·2817	1·366

Construct the quadratic lagrange interpolating polynomial that fits the data. Hence, find $f(\frac{\pi}{12})$. Compare with the exact value.

(b) Derive the Newton's forward difference formula using the operator relations. [7]

Q.8 Prove the following:

(a) $\int J_3(x)dx=c-J_2(x)-\frac{2}{x}J_1(x)$, where c is arbitrary constant. [6]

(b) $$\int_{-1}^{1}x^2P_{n+1}(x)P_{n-1}(x)dx=\frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}$$ [8]

Q.9 Write shorts notes on ***any two*** of the following: [7x2=14]

(a) Truncation error of the Taylor's series method.

(b) Laplace's equation.

(c) Cauchy-Reimann equation.

© Apna BEU
CONFIDENTIAL DOCUMENT EDU-159 @ 03-10-2023 13:01

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play